Network Scanning with Zenmap

Week 2 – Project Module 5 (W2-PM5) Report

Platform: Windows | Tools: Zenmap, Nmap, Windows CMD | Scope: my own local network only

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1. Project Objective

The objective of this project is to use Zenmap, the graphical interface of Nmap, to discover live hosts on my local area network (LAN). The project covers identifying my own IP address and subnet, running a ping scan, counting live hosts, recording their IP and MAC addresses, and exporting the network topology as a PDF.

Specific goals:

- Download and install Zenmap on Windows.
- Find the local IP address and LAN subnet using CMD.
- Run a ping scan to find live hosts in the subnet.
- Count the live hosts and record their IP addresses.
- Record MAC addresses where they are available.
- Save the network topology diagram as a PDF.

2. Introduction to Zenmap and Nmap

Nmap (Network Mapper) is a free, open-source tool for network discovery and security auditing. Zenmap is its official graphical front end, which makes scanning easier for beginners by providing scan profiles, organized result tabs and a topology viewer.

Zenmap provides these views of a scan:

- **Nmap Output:** the raw text output of the scan.
- **Ports / Hosts:** the list of discovered hosts and their details.
- **Topology:** a graphical map of the discovered network.
- **Host Details and Scans:** details per host and a history of scans run.

Ping scan

A ping scan (host discovery) checks which devices respond on the network without scanning their ports. It is quick and low-impact, which makes it a common first step in network reconnaissance.

3. Tools and Technologies Used

Item	Details
Operating system	Windows (laptop)
Scanner	Zenmap (GUI for Nmap)
Command line	Windows CMD (ipconfig)
Scan profile	Ping scan
Output	Screenshots and topology exported as PDF
Documentation	GitHub repository, LinkedIn post

4. Ethical and Legal Notice

All scans in this project were performed only on my own network, or a network I am authorized to test. Scanning networks without permission may be illegal and against the policies of network owners. No ports were scanned, and no systems were accessed or modified.

5. Task 1 – Download and Install Zenmap

Steps performed:

1. Opened the official Nmap website and went to the download page.
2. Downloaded the Windows installer.
3. Ran the downloaded .exe file.
4. Followed the installation wizard, then clicked Install and Finish.
5. Opened Zenmap from the Windows Start menu.

Nmap/Zenmap version installed: [fill in]

[Insert screenshot: Zenmap installation / Zenmap opened]

6. Task 2 – Find Local IP Address and LAN Subnet

Steps performed:

1. Pressed Windows + R, typed cmd and pressed Enter.
2. Ran the following command:

```
ipconfig
```

3. Looked under the active Wi-Fi or Ethernet adapter for the IPv4 address, subnet mask and default gateway.

Item	My value
IPv4 address	[fill in]
Subnet mask	[fill in]
Default gateway	[fill in]
LAN subnet (CIDR)	[fill in, e.g. a.b.c.0/24 for mask 255.255.255.0]

[Insert screenshot: CMD ipconfig output]

7. Task 3 – Find Live Hosts in the Subnet

Steps performed:

1. Launched Zenmap.
2. Entered my subnet in the Target field: [your subnet]
3. Selected the **Ping scan** profile.
4. Clicked **Scan** and waited for it to finish.
5. Reviewed the results in the Hosts, Nmap Output and Topology tabs.

Command generated by Zenmap: [copy the command shown, usually similar to nmap -sn <subnet>]

Scan duration: [fill in]

[Insert screenshot: Zenmap ping scan with command and results]

8. Task 4 – Count of Live Hosts

After the scan finished, I opened the Hosts tab and counted the discovered live hosts, including my own PC where it appeared.

Total live hosts: [actual number from your scan]

[Insert screenshot: Hosts tab showing live hosts]

9. Task 5 – IP Addresses of Live Hosts

I used the Nmap Output tab to identify the hosts reported as up.

No.	IP address	Notes (e.g. router, my laptop, phone)
1	[fill in]	[fill in]
2	[fill in]	[fill in]
3	[fill in]	[fill in]
4	[fill in]	[fill in]

(Add or remove rows to match your results.)

[Insert screenshot: Nmap Output with host IP addresses]

10. Task 6 – MAC Addresses of Live Hosts

Method 1 – From Zenmap: On a local LAN, the Nmap Output tab may show a line starting with MAC Address for hosts discovered through local network discovery.

Method 2 – Using CMD for my own laptop:

```
ipconfig /all
```

The value under Physical Address is my laptop's MAC address.

IP address	MAC address	Vendor shown (if any)
[fill in]	[fill in or "not shown"]	[fill in]
[fill in]	[fill in or "not shown"]	[fill in]

Limitation: Zenmap may not display the MAC address of every host, and Windows does not always reveal MAC addresses of remote devices. Any MAC address not displayed is marked "not shown" above and was not guessed.

[Insert screenshot: MAC addresses in Nmap Output / ipconfig /all]

11. Task 7 – Save Network Topology as PDF

Steps performed:

1. After the scan, opened the **Topology** tab.

2. Enabled the legend where available.
3. Clicked **Save Graphic**.
4. Chose PDF as the format and Desktop as the location.
5. Saved the file as Zenmap_Network_Topology.pdf.

[Insert screenshot: Topology tab]

[Insert screenshot: Saved Zenmap_Network_Topology.pdf]

Topology observation: [describe the diagram, for example the gateway at the center with devices connected to it].

12. Summary of Results

Task	Description	Result
1	Install Zenmap	[Completed / version]
2	Local IP and subnet	[IP and subnet]
3	Ping scan for live hosts	[Completed]
4	Number of live hosts	[number]
5	IP addresses of live hosts	[list]
6	MAC addresses	[list / limitations]
7	Topology saved as PDF	[filename]

13. Security Implications

Host discovery shows how visible devices are on a network. Attackers who gain access to a network can run the same scan to find targets. Good practices that reduce risk include:

- Keeping an inventory of the devices that should be on the network
- Checking regularly for unknown or unexpected devices
- Securing the router with a strong password and updated firmware
- Using WPA2/WPA3 and a strong Wi-Fi password
- Separating guest and IoT devices from main devices
- Using firewalls and network monitoring

14. Learning Outcomes

- Learned how to install and use Zenmap.
- Learned to find an IP address, subnet mask and gateway with ipconfig, and to express a subnet in CIDR notation.
- Understood how a ping scan finds live hosts without scanning ports.
- Learned to read results in the Hosts, Nmap Output and Topology tabs.
- Understood the limits of MAC address discovery and the importance of reporting results honestly.
- Practiced documenting practical work with screenshots, GitHub and a written report.

15. Conclusion

This project showed how Zenmap and Nmap can be used to discover live hosts on a local network, identify their IP and, where available, MAC addresses, and visualize the network as a topology diagram. A ping scan is a fast and lightweight way to understand what is connected to a network, and it is a foundation for deeper security assessments. [Add one or two sentences about what you found on your own network.]

16. Project Files

```
Week2-PM5-Zenmap-Network-Scanning/
├── README.md
├── Report/
│   └── Zenmap_Network_Scanning_Report.pdf
├── Screenshots/
│   ├── ipconfig.png
│   ├── ping_scan.png
│   ├── live_hosts.png
│   ├── mac_addresses.png
│   └── topology.png
└── Results/
    └── Zenmap_Network_Topology.pdf
```

GitHub repository: [Paste your repository link]

17. References

- Nmap official website and Zenmap documentation (nmap.org)
- Nmap Reference Guide: host discovery options
- Course material: W2-PM5 – Network Scanning with Zenmap